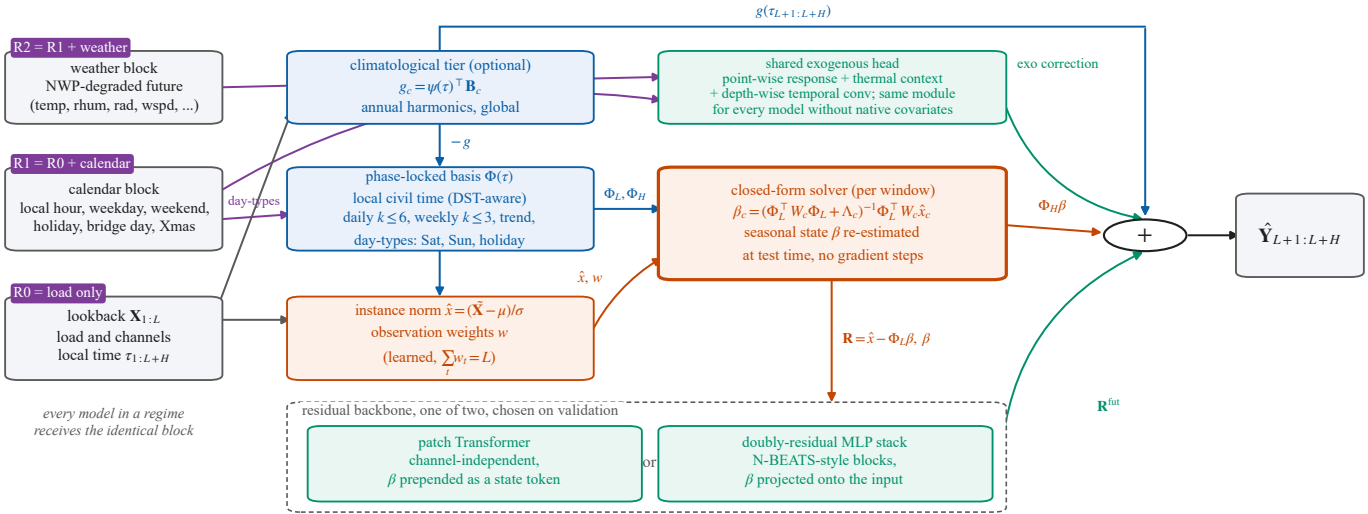

$g(\tau_{L+1:L+H})$
R2 = R1 + weather
weather block
NWP-degraded future
(temp, rhum, rad, wspd, ...)
R1 = R0 + calendar
calendar block
local hour, weekday, weekend,
holiday, bridge day, Xmas
day-types
R0 = load only
lookback $\mathbf{X}_{1:L}$
load and channels
local time $\tau_{1:L+H}$
every model in a regime
receives the identical block
climatological tier (optional)
$g_c = \psi(\tau)^\top \mathbf{B}_c$
annual harmonics, global
$-\,g$
phase-locked basis $\Phi(\tau)$
local civil time (DST-aware)
daily $k \le 6$, weekly $k \le 3$, trend,
day-types: Sat, Sun, holiday
instance norm $\hat{x} = (\mathbf{X} - \mu)/\sigma$
observation weights $w$
(learned, $\sum_t w_t = L$)
shared exogenous head
point-wise response + thermal context
+ depth-wise temporal conv; same module
for every model without native covariates
exo correction
$\Phi_L, \Phi_H$
$\hat{x}$, $w$
closed-form solver (per window)
$\beta_c = (\Phi_L^\top W_c \Phi_L + \Lambda_c)^{-1} \Phi_L^\top W_c \hat{x}_c$
seasonal state $\beta$ re-estimated
at test time, no gradient steps
$\Phi_H \beta$
+
$\hat{\mathbf{Y}}_{L+1:L+H}$
$\mathbf{R} = \hat{x} - \Phi_L \beta$, $\beta$
$\mathbf{R}^{\text{fut}}$
residual backbone, one of two, chosen on validation
patch Transformer
channel-independent,
$\beta$ prepended as a state token
or
doubly-residual MLP stack
N-BEATS-style blocks,
$\beta$ projected onto the input